

Lecture 03: Repeated Games and the Theory of Cooperation

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Outline of MAAI

- 1 Basic Game Theory and Nash Equilibrium
- 2 Potential Games and Best Respond Dynamics
- 3 Repeated Games and the Theory of Cooperation
- 4 Minimax Games
- 5 General-Sum Games and L-H Algorithms
- 6 Single-agent Reinforcement Learning and MDP
- 7 Learning Stochastic Games
- 8 Non-regret Learning and Correlated Equilibrium
- 9 Counterfactual Regret Minimisation
- 10 Learning with a Population of Agents

- Subject to change

Key Preferences

- Martin J. Osborne. *An Introduction to Game Theory*. Oxford University Press, 2004
- Robert Axelrod and William D Hamilton. “The evolution of cooperation”. In: *science* 211.4489 (1981), pp. 1390–1396
- Robert Axelrod. “The emergence of cooperation among egoists”. In: *American political science review* 75.2 (1981), pp. 306–318

Outline

1 Motivation

2 Repeated Games

- Framework and Definition
- The “Threat” of Punishment
- Discounting Future Payoffs
- Equilibria in Repeated Games
- Open Questions and Formalisation
- Grim Trigger Strategys
- Limited Punishment Strategy
- Tit-for-tat Strateg
- The Iterated Prisoners’ Dilemma

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Canonical Matrix Game Social Dilemmas¹

	C	D
C	R, R	S, T
D	T, S	P, P

- ① R : Reward of mutual cooperation.
- ② P : Punishment from mutual defection.
- ③ S : Sucker outcome when cooperating with a defector.
- ④ T : Temptation outcome from defecting against a cooperator.

- **Social dilemmas**: individual incentives vs. collective interests.
- In **Chicken**, agents defect out of greed; in **Stag Hunt**, out of fear of non-cooperation; and in **Prisoner's Dilemma**, out of both greed and fear.

	Chicken	
	C	D
C	3, 3	1, 4
D	4, 1	0, 0

	Stag Hunt	
	C	D
C	4, 4	0, 3
D	3, 0	1, 1

	Prisoner's Dilemma	
	C	D
C	3, 3	0, 4
D	4, 0	1, 1

¹Joel Z Leibo et al. "Multi-agent reinforcement learning in sequential social dilemmas". In: *Proceedings of AAMAS*. 2017.

Social Dilemma Inequalities

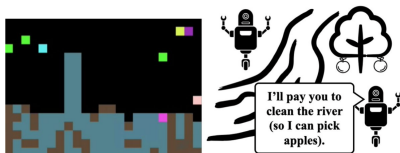
A matrix game is a **social dilemma** if the following inequalities hold²:

- ① $R > P$: Mutual cooperation is preferred to mutual defection.
- ② $R > S$: Mutual cooperation is preferred to being exploited by a defector.
- ③ $2R > T + S$: Mutual cooperation is preferred to an equal probability of unilateral cooperation and defection.
- ④ Either:
 - **Greed**: $T > R$: Exploiting a cooperator is preferred over mutual cooperation, or
 - **Fear**: $P > S$: Mutual defection is preferred over being exploited.

²Joel Z Leibo et al. "Multi-agent reinforcement learning in sequential social dilemmas". In: *Proceedings of AAMAS*. 2017.

Social Dilemma Solved by Social Contracts³

- In the **Cleanup domain** game, agents earn rewards by picking apples that only grow if the nearby river is unpolluted.
- A **pro-social solution** requires cooperation:
 - One agent cleans the river; The other picks apples.
- **Self-interested agents** cannot sustain this:
 - Cleaning offers no immediate reward, so selfish agents focus on picking apples.
- This creates a **social dilemma** between cooperation and self-interest.



Contracts: One agent *pays* the other to clean the river, enabling coordinated policies that maximise total rewards for both agents.

³Andreas Haupt et al. "Formal contracts mitigate social dilemmas in multi-agent reinforcement learning". In: *Autonomous Agents and Multi-Agent Systems* 38.2 (2024), pp. 1–38.

Social Dilemma Solved by Social Contracts⁴

Contract:

Defecting agents are fined 1.5 units by the other. If both defect, the fines cancel.

Table 1: Prisoner's Dilemma with and without a contract where a defector transfers 1.5 reward to a cooperator.

(a) Without Contract

	C	D
C	-1, -1	-3, 0
D	0, -3	-2, -2

(b) With Contract

	C	D
C	-1, -1	-1.5, -1.5
D	-1.5, -1.5	-2, -2

With this contract in force, cooperating becomes a dominant action for both players - In this modified game, cooperation becomes dominant, making (C, C) the only Nash equilibrium.

⁴Andreas Haupt et al. "Formal contracts mitigate social dilemmas in multi-agent reinforcement learning". In: *Autonomous Agents and Multi-Agent Systems* 38.2 (2024), pp. 1–38.

Repeated Games: Motivation

- Many strategic interactions in life are ongoing, involving **repeated engagements** with the same individuals.
- Despite opportunities for short-term exploitation, people often act **cooperatively**.

Examples include:

- Looking after a neighbour's house while they are away.
- Lending money to friends in temporary need.
- Answering questions in online forums (e.g., Stack Overflow, Yahoo! Answers).

The theory of repeated games provides a framework to study such behaviour.

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Definition and Framework

- Let us consider a **repeated game** involving the repetition of a normal-form (simultaneous-move) game over multiple time periods.
- The game played in each period is called the **stage game**.
- Example: Players may repeatedly engage in a Prisoner's Dilemma for several rounds.

Payoffs in the Prisoner's Dilemma:

Player 1/Player 2	C (Cooperate)	D (Defect)
Player 1: C	(2, 2)	(0, 3)
Player 1: D	(3, 0)	(1, 1)

The “Threat” of Punishment

- Equilibria in repeated games can differ from those of the associated stage game due to:
 - The ability to condition strategies on the history of play, resulting in many new strategies;
 - A player may be deterred from exploiting her short-term advantage by the threat of punishment that reduces her long-term payoff.

Example: Repeated Prisoner's Dilemma

- Recall the Prisoner's Dilemma has a unique Nash equilibrium, in which each player chooses D.

Player 1/Player 2	C (Cooperate)	D (Defect)
Player 1: C	(2, 2)	(0, 3)
Player 1: D	(3, 0)	(1, 1)

- Suppose we repeatedly play it against **Grim Trigger Strategy**:
 - Cooperate (C) as long as the other player cooperates.
 - If the other player defects (D), defect in all future rounds.
 - Incentives to Cooperate:
 - Cooperation yields a long-term payoff stream of $(2, 2, 2, \dots)$.
 - Defection results in $(3, 1, 1, \dots)$.
 - Cooperation is preferable if future payoffs are **sufficiently valued**.
- How to define and measure that?

Discounting Future Payoffs

- Future payoffs are **discounted** because:
 - The game may end with probability $1 - \delta$.
 - Future rewards are less valuable due to inflation or opportunity costs.
- Using discount factor $\delta \in (0, 1)$:
 - Payoff in period t : $\delta^t u_t$.
- For an **Infinitely Repeated Game**, player i 's total utility for history action $h = (a^0, a^1, a^2, \dots)$ is:

$$u_i(h) = \sum_{k=0}^{\infty} \delta^k u_i(a^k),$$

where $a^k = (a_k^1, a_k^2)$ is the set of actions for all the two players at time k .

Some Conclusions about Equilibria in Repeated Games

Cooperation as a Nash Equilibrium:

- Cooperation can form a Nash equilibrium if players are sufficiently patient (δ is high).
- Punishment deters defection.

Defection as a Nash Equilibrium:

- Mutual defection is also a Nash equilibrium.
- If one player defects unconditionally, the other has no incentive to cooperate.

Open Questions

- How patient must players be (i.e., what value of δ is required) for cooperation to be a Nash equilibrium?
- What other outcomes and strategies can be supported as equilibria?

Formal Definition of Repeated Game

Repeated Game:

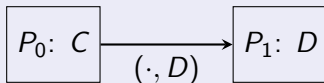
- A stage game $G = (N, A, u)$ is repeated over multiple periods.
- Players have perfect information about past actions but not current simultaneous actions.
- Utility in a finitely repeated game is the sum of utilities across all rounds.

Grim Trigger Strategy

Definition: Grim Trigger Strategy

Cooperate until the opponent defects, then defect forever.

- Strategy has two states:
 - 1 P_0 : Cooperate (C).
 - 2 P_1 : Defect (D) — triggered if the opponent defects.
- Initially, the state is P_0 ; Transition from P_0 to P_1 is permanent if defection occurs.



Utility of Grim Trigger Strategy

Player 1/Player 2	C (Cooperate)	D (Defect)
Player 1: C	(2, 2)	(0, 3)
Player 1: D	(3, 0)	(1, 1)

- Suppose both players adopt the Grim Trigger strategy:

$$u_2(h) = 2 + 2\delta + 2\delta^2 + \dots = \frac{2}{1 - \delta}$$

- If one deviates, payoffs become $(3, 1, 1, \dots)$:

$$u_2'(h) = 3 + \delta + \delta^2 + \dots = \frac{3 - 2\delta}{1 - \delta}.$$

Condition for Nash Equilibrium (Both Taking Grim Trigger)

No incentive to deviate if:

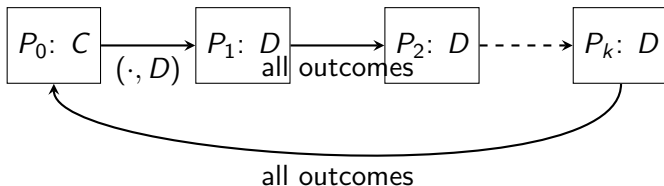
$$\frac{2}{1 - \delta} \geq \frac{3 - 2\delta}{1 - \delta} \implies \delta \geq \frac{1}{2}.$$

Limited Punishment Strategy: Definition

- Punishes deviations for a fixed number of k periods.
- Transition:

$$P_0 \xrightarrow{D} P_1 \xrightarrow{k \text{ periods}} P_0.$$

- After punishment, reverts to cooperation regardless of past behaviour.



Limited Punishment Strategy: Analysis

- Suppose Player 2 chooses to deviate, say D in the first period.
- Then Player 1 chooses D in each of the next k periods, regardless of Player 2's choices.

- Player 2 also should choose D in these periods, why?

Player 1: C D D $\dots$ D

Player 2: D D D $\dots$ D

- In the $(k + 1)$ th period after the deviation, Player 1 switches back to C (regardless of Player 2's behaviour previously).

- Player 1: C D D $\dots$ D C C $\dots$ C

Player 2: D D D $\dots$ D C C $\dots$ C

- Again, Player 2 faces precisely the same situation at the beginning. Let us say she is back to being "nice" again.
- So if the deviation increases her payoff, it increases her payoff during the first $k + 1$ periods, which is:

$$3 + \delta + \delta^2 + \delta^3 + \dots + \delta^k = 2 + \frac{1 - \delta^{k+1}}{1 - \delta}$$

Why does the equality hold?

Limited Punishment Strategy: Condition for NE

- If Player 2 does not deviate during the first $k + 1$ periods:
 - Player 1: C C C $\dots$ C
Player 2: C C C $\dots$ C
- Her payoff (utility) is:

$$2 + 2\delta + 2\delta^2 + \dots + 2\delta^k = \frac{2(1 - \delta^{k+1})}{1 - \delta}$$

- Thus, she cannot increase her payoff by deviating if and only if:

$$\frac{2(1 - \delta^{k+1})}{1 - \delta} \geq 2 + \frac{1 - \delta^{k+1}}{1 - \delta}$$

- Which leads to:

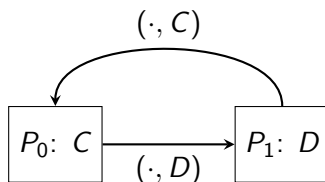
$$\delta^{k+1} - 2\delta + 1 \leq 0$$

Limited Punishment Strategy: Implications

- No-deviation condition $\delta^{k+1} - 2\delta + 1 \leq 0$ suggests that:
 - If $k = 1$: Then no value of δ less than 1 satisfies the inequality:
One period of punishment is not severe enough to discourage deviation, no matter how patient the players are.
 - If $k = 2$: Then the inequality is satisfied for $\delta \geq 0.62$.
 - If $k = 3$: It is satisfied for $\delta \geq 0.55$.
- As k increases, the lower bound on δ approaches $1/2$, the lower bound for the grim strategy.

Tit-for-tat Strategy

- In the strategy **tit-for-tat**, the length of the punishment depends on the behaviour of the player being punished:
 - If she continues to choose D, then *tit-for-tat* continues to do so.
 - If she reverts to C, then *tit-for-tat* reverts to C also.



Condition that Tit-for-tat Strategy is a Nash Equilibrium

- Suppose that Player 1 adheres to this strategy starting with C.
- Then, if Player 2 can gain by deviating, then she can gain by choosing D in the first period. Why?
 - Player 1: C
Player 2: D
- If she does so, Player 1 chooses D in the second period and continues to choose D until Player 2 reverts to C.
 - Player 1: C D
Player 2: D
- Thus Player 2 has two options:
 - Revert to C, in which case, in the next period, she faces the same situation as she did at the start of the game.
 - Continue to choose D, then Player 1 will continue to do so too.

Tit-for-tat Strategy

- If she alternates between D and C, then her stream of payoffs is $(3, 0, 3, 0, \dots)$ with:

$$\frac{3}{(1 - \delta)(1 + \delta)}.$$

- If she chooses D in every period, her stream of payoffs is $(3, 1, 1, 1, \dots)$ with utility:

$$\frac{3 - 2\delta}{1 - \delta}.$$

- Since Player 2 utility of adhering to the strategy **tit-for-tat** is:

$$\frac{2}{1 - \delta}.$$

- Thus **tit-for-tat** is a best response to **tit-for-tat** if and only if:

$$\frac{2}{1 - \delta} \geq \frac{3 - 2\delta}{1 - \delta} \quad \text{and} \quad \frac{2}{1 - \delta} \geq \frac{3}{(1 - \delta)(1 + \delta)}$$

- Both of these conditions are equivalent to: $\delta \geq \frac{1}{2}$.

The Iterated Prisoners' Dilemma (IPD) Tournament⁵

- Robert Axelrod and William D Hamilton. “The evolution of cooperation”. In: *science* 211.4489 (1981), pp. 1390–1396 reported a tournament with:
 - The *N*-step Prisoners' Dilemma (with *N* fixed).
- Researchers all over the world were invited to devise computer strategies to compete:
 - Programs that were entered varied widely.
- When playing with many players, each with different strategies:
 - Greedy strategies tended to perform very poorly.
 - More altruistic strategies performed better.
- The winning deterministic strategy was tit for tat.

⁵Robert Axelrod and William D Hamilton. “The evolution of cooperation”. In: *science* 211.4489 (1981), pp. 1390–1396.

The Iterated Prisoners' Dilemma (IPD) Tournament

- If the discount parameter $\delta > \frac{T-R}{T-P}$ is sufficiently high, there is no dominant strategy.
- *No better strategy can invade* (do better than) *tit for tat* population *iff*:

$$\delta > \max \left(\frac{T-R}{T-P}, \frac{T-R}{R-S} \right)$$

- A small group of strategies that can maximally discriminate bad/good ones (act accordingly), such as *tit for tat*, can invade (do better than) an **ALL D** population. This connects to *evolutionarily stable strategies*, to be explained later.

P_1/P_2	C	D
C	R, R	S, T
D	T, S	P, P

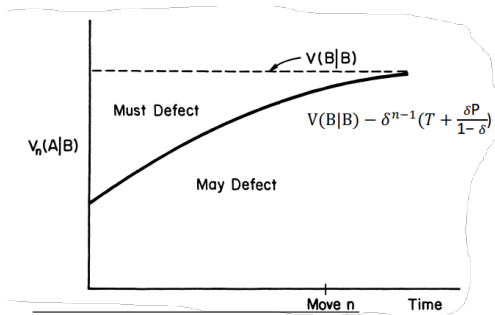
Axelrod, Robert. *"The emergence of cooperation among egoists."*
American political science review 75.2 (1981): 306-318.

The Iterated Prisoners' Dilemma (IPD) Tournament⁶

- A strategy B is NE iff it defects immediately whenever in stage n the other player's cumulative payoff (with strategy A):

$$V_n(A|B) > V(B|B) - \delta^{n-1} \left(T + \frac{\delta P}{1 - \delta} \right)$$

- Suppose strategy A exploits B by obtaining $V_n(A|B)$ in the first n steps:
 - B must defect in step $n \leq \ln \frac{V - V_n}{T + \frac{\delta P}{1 - \delta}} + 1$ in order to do better.



P_1/P_2	C	D
C	R, R	S, T
D	T, S	P, P

- $V(B|B)$: Value of strategy B when played against B , i.e., $V(C|C) = R$.
- $V_n(A|B)$: Value of strategy A when played against B for n steps.

⁶Robert Axelrod. "The emergence of cooperation among egoists". In: *American political science review* 75.2 (1981), pp. 306–318.

Conclusions

Key Topics:

- **Social Dilemmas:** Individual incentives vs. collective interests.
- **Repeated Games:** Studying cooperation over time.
- **Strategies:**
 - Grim Trigger: Punishes defection permanently.
 - Limited Punishment: Temporary penalties for deviation.
 - Tit-for-Tat: Mimics the opponent's actions.
- **Iterated Prisoner's Dilemma:** Demonstrates the emergence of cooperation through strategies like Tit-for-Tat.

Future Topics:

- Evolutionarily Stable Strategies (ESS) and their connection to Nash equilibria.
- Markov Games: Extending repeated games to add multiple states (stage games).

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References I

- [AH81] Robert Axelrod and William D Hamilton. “The evolution of cooperation”. In: *science* 211.4489 (1981), pp. 1390–1396.
- [Axe81] Robert Axelrod. “The emergence of cooperation among egoists”. In: *American political science review* 75.2 (1981), pp. 306–318.
- [Hau+24] Andreas Haupt et al. “Formal contracts mitigate social dilemmas in multi-agent reinforcement learning”. In: *Autonomous Agents and Multi-Agent Systems* 38.2 (2024), pp. 1–38.
- [Lei+17] Joel Z Leibo et al. “Multi-agent reinforcement learning in sequential social dilemmas”. In: *Proceedings of AAMAS*. 2017.
- [Osb04] Martin J. Osborne. *An Introduction to Game Theory*. Oxford University Press, 2004.